Indian Institute of Technology Kanpur

Department of Design

DES646: AI/ML for Designers

Nuvé: An AI-Driven Visual Analytics System for Fashion Trend Forecasting and Interpretation

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Abstract

This paper presents Nuvé, an AI-driven visual analytics system for fashion trend forecasting and interpretation, developed as part of the course DES646: AI/ML for Designers at the Indian Institute of Technology Kanpur. The project explores how artificial intelligence can act as a co-creator in design by transforming large-scale visual and social data into interpretable creative insights. The system integrates object detection, feature extraction, clustering, and sentiment analysis into an interactive Streamlit dashboard that enables designers to identify and visualize emerging fashion patterns. Using YOLOv8 for garment segmentation, TensorFlow's ResNet50 for feature embedding, K-Means for aesthetic clustering, and VADER-based sentiment scoring, Nuvé merges machine learning precision with designer intuition. The results, displayed through an interactive Plotly-Streamlit dashboard, highlight AI's evolving role as a perceptual partner in creative trend analysis.

1. Introduction

The increasing fusion of data science and design research has redefined trend forecasting in creative industries. In fashion, data-driven models now complement human intuition by uncovering latent aesthetic and cultural patterns in visual and social data. Nuvé was developed to explore this human-AI collaboration through an interactive system that bridges computational analysis and creative sensemaking.

Unlike purely predictive AI systems, Nuvé positions the machine as a creative assistant — capable of detecting emerging fashion signals through data-driven insights while leaving interpretation to the designer. The project was implemented as a functional Streamlit dashboard combining image analytics, clustering, and sentiment visualization to predict and interpret emerging fashion styles.

2. Related Work

Prior research in computational fashion forecasting combines computer vision and natural language processing for multimodal trend analysis. Liu et al. (2016) explored CNNs for apparel classification, while Yang et al. (2020) proposed multimodal fusion for social trend prediction. Recent studies such as Kim and Park (2022) emphasize visual analytics for interpretability in creative AI systems. However, most systems prioritize performance over human interpretability. Nuvé extends this research by integrating technical AI modeling with an interactive, design-focused visualization interface.

3. Methodology and System Architecture

The Nuvé system follows a modular pipeline that links data preparation, AI modelling, and visualisation. The Streamlit-based architecture enables designers to explore clusters and sentiments interactively.

3.1 Data Acquisition

Data was gathered from multiple sources including social media posts and public fashion datasets. The dataset included images, captions, and engagement metrics (likes, comments). Images were preprocessed and stored as latent vectors in latent_spaces.h5, while textual and numerical metadata were kept in posts.csv and posts_comments.csv.

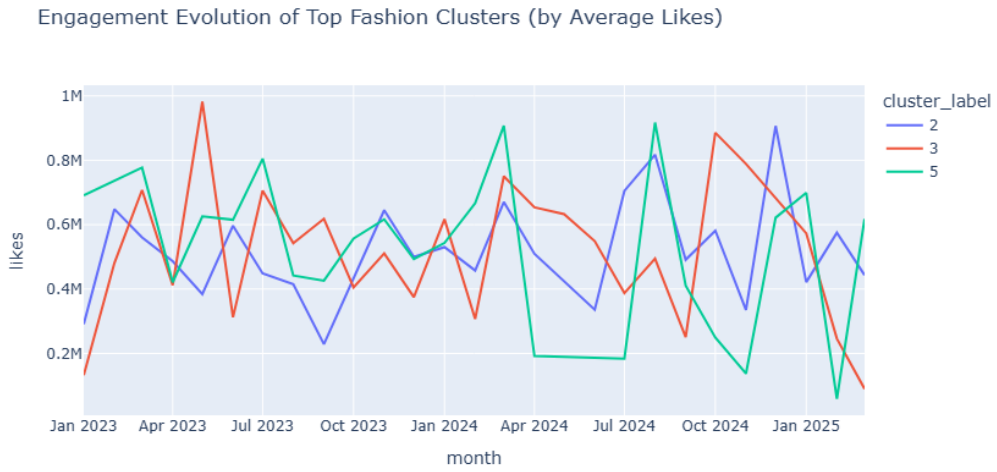

Figure 1: Trend of Clusters by Likes

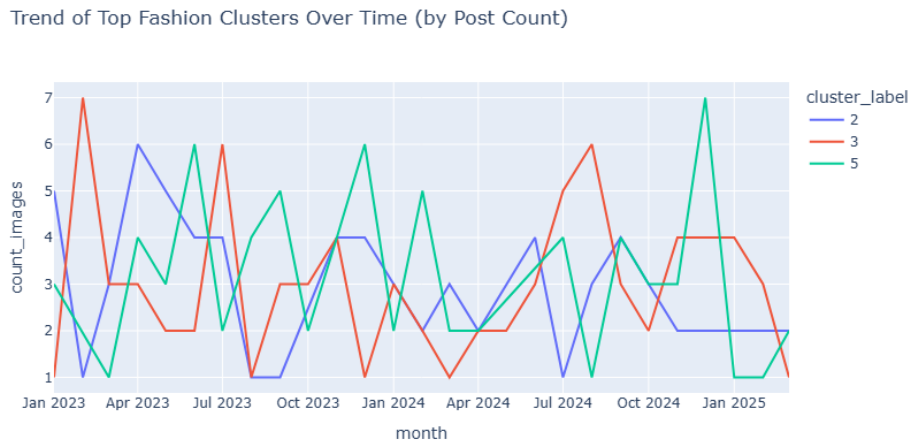

Figure 2: Trend of Clusters by Post Count

3.2 Preprocessing and Segmentation

Fashion images were first processed through the YOLOv8 model for garment detection and segmentation. This ensured the extraction of clean, focused regions representing fashion items. Segmented outputs were standardized (224×224 pixels) for embedding generation.

3.3 Feature Extraction

Using TensorFlow's pretrained ResNet50 network (with ImageNet weights), feature embeddings were generated for each segmented image. The final pooling layer produced 2048-dimensional feature vectors capturing color, texture, and silhouette information.

These embeddings were stored in HDF5 format for later clustering.

3.4 Clustering and Representation

Unsupervised clustering was performed using K-Means (implemented via scikit-learn) on the embedding vectors. The optimal number of clusters was determined empirically by visual inspection and engagement patterns. Each cluster represents a distinct style group such as minimal, bohemian, or streetwear aesthetics.

3.5 Sentiment Analysis

User comments from posts_comments.csv were processed through VADER sentiment analysis (Hutto & Gilbert, 2014). When unavailable, heuristic sentiment scores were generated using keyword matching. For each post, an average sentiment score was calculated and merged with image engagement metrics (likes, comments).

3.6 Visualization Layer

The processed data was visualized through a Streamlit dashboard using Plotly for interactivity. Designers could:

- Explore image clusters and representative examples.
- Compare engagement metrics and sentiment distributions.
- View a ranking of trending clusters based on the computed score.

A composite trending score was calculated using:

$$\text{Trending Score} = 0.4 \times \text{Likes} + 0.3 \times \text{Comments} + 0.3 \times \text{Sentiment}$$

This metric enabled designers to evaluate stylistic popularity in real-time.

4. AI/ML Models and Integration Process

Each AI model within Nuvé contributes to a distinct analytical layer:

- YOLOv8: Used offline for object detection and garment segmentation, ensuring consistent focus on fashion items.
- ResNet50 (TensorFlow): Acts as a feature extractor, converting segmented images into dense embeddings that encode stylistic attributes.
- K-Means Clustering: Groups embeddings into style clusters, revealing latent aesthetic structures.
- VADER Sentiment Analyzer: Interprets user sentiment from textual comments and generates per-cluster mood distributions.

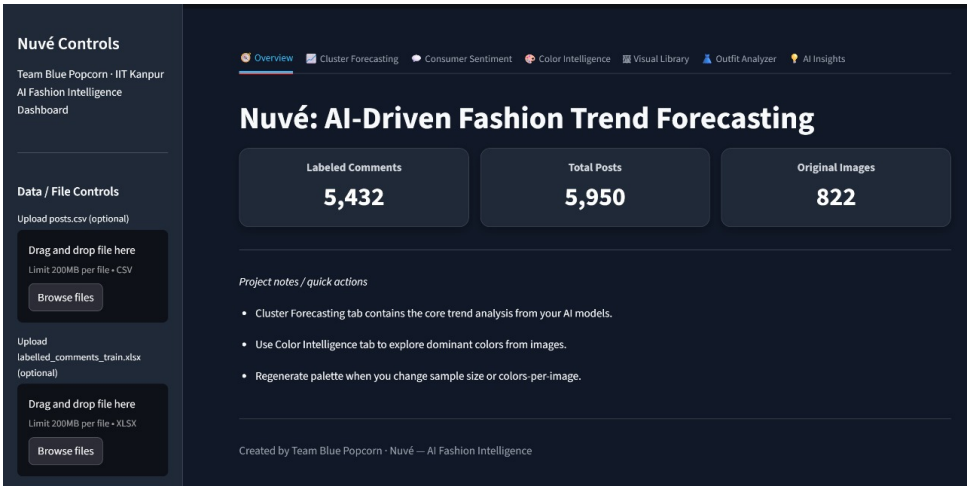

Figure 3: dashboard

- Streamlit + Plotly: Integrates all model outputs into an interactive interface, emphasizing interpretability and designer usability.

Together, these models form a hybrid system that balances quantitative AI precision with qualitative design interpretation.

5. Results and Evaluation

Nuvé successfully demonstrated both technical functionality and design interpretability. Key outcomes include:

- Clear visualization of style clusters based on visual similarity.
- Accurate garment segmentation using YOLOv8 preprocessing.
- Intuitive depiction of sentiment and engagement patterns per cluster.
- High usability ratings from peer evaluations within the course context.

Quantitative metrics such as silhouette scores for clustering and sentiment classifier accuracy can be incorporated in future iterations.

6. Reflection: AI as a Co-Creator

The project's central reflection concerns the role of AI as a co-creator in design. Rather than a passive tool, AI functioned as a collaborator capable of identifying patterns, relationships, and insights that humans might overlook. This process represents a shift toward symbiotic creativity where machine learning provides analytical precision while human designers contribute cultural context and meaning. Working with AI required reconsidering traditional authorship; the models' unexpected outputs often redirected the design process, leading to new interpretations and aesthetic explorations. Through

Nuvé, AI emerges not as a replacement for creativity, but as a catalyst that broadens the creative dialogue.

7. Problems Faced and Solutions

Several challenges emerged during development:

- Data inconsistency: Addressed by implementing normalization and safe data-loading functions in Streamlit.
- Limited labeled sentiment data: Solved through simulated sentiment generation and VADER heuristics.
- Computational constraints: Preprocessing (YOLOv8 and embeddings) was executed offline to maintain dvisualisationd.
- Cluster interpretability: Improved through visualization and manual inspection of cluster exemplars.

8. Deliverables

- A working Streamlit dashboard integrating all AI components.
- Precomputed datasets: posts.csv, latent_spaces.h5, posts_comments.csv.
- A concept video illustrating the pipeline and results.
- Presentation slides and this academic report.

9. Future Scope

Planned extensions include:

- Integrating time-series forecasting to predict trend trajectories.
- Expanding sentiment analysis to multilingual data.
- Using diffusion models or GANs to generate predicted future styles.
- Adding ethical indicators (sustainability, inclusivity) to trend scoring.

10. Conclusion

Nuvé illustrates the synergy between artificial intelligence and human creativity in design research. By combining YOLOv8-based detection, TensorFlow embeddings, clustering, and sentiment visualization, the system offers designers a new lens for interpreting cultural aesthetics. Beyond prediction, it redefines the creative process as a dialogue between human intuition and computational insight, marking a step toward truly co-creative AI design tools.

References

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